San Francisco, California

San Francisco's storm failures have rarely been missing warnings. On 31 December 2022 the forecast was there, and the city waited to see whether the warning was serious. This winter, a strong El Niño makes a wet one likely. The single most important step is to commit in writing, before November, that a National Weather Service flood watch on its own opens the emergency centre, puts the flood barriers up, and opens storm-night shelter. After that, the people to reach first are those stuck on upper floors when the power and elevators fail.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
San Francisco County, CA (inferred; the city and county share one boundary, ~810,000–830,000 people)	None published for the US, California, or San Francisco	n/a here; global 6-month figure 239,000 ± 11,000	Atmospheric-river nights: exposure for ~3,400 unsheltered people, multi-day outages that strand people on upper floors, carbon monoxide, sewer-backup flooding, falling trees, and bluff erosion at South Ocean Beach

EXECUTIVE SUMMARY — AT A GLANCE



37.79°N 122.41°W, United States of America · Natural Earth

- **The risk.** Storm failures here have rarely been missing warnings: on 31 December 2022 the forecast was there and the city waited, and about 3,400 unsheltered people are exposed on atmospheric-river nights.
- **First action.** DEM, SFPUC, Public Works, and HSH commit in writing by 15 November that a Flood Watch or AR3+ forecast alone opens the emergency centre, deploys the barriers, and activates storm-night shelter.
- **The money.** $253k floor, $605k target, $2.58M ceiling; the city does not need donor money to build shelter, but a donor can buy a lower trigger for opening it.
- **Do not.** Do not clear encampments or tow lived-in vehicles around a storm.

What to do	**9** actions	p. 2
Who does what	RACI chart	p. 12
Where money does the most	funding	p. 12
Do not do this	**3** items	p. 14
What has happened here	the record	p. 15
How this was produced	methodology	p. 17
References	sources	p. 19

* Suggested formulation: check with a health worker before sharing.

12/12 recommendations checked. 0 confirmed outright, 10 corrected, 1 unverified (a flood line in the city's reserves), and 1 contraindicated (binding municipal parametric flood cover this winter). Three further practices are flagged “do not do”. Exposure figure: Climate Impact Lab, September 2026, pre-peer-review, heat only. No figure exists at this resolution, and heat is not San Francisco's winter hazard, so no number is borrowed. **The funding target fell 29% in verification**, from $855k to $605k, because a pooled fund cannot be restricted to storms. Several verifiers ran out of web-search budget and checked by fetching primary pages; claims they could not reach are marked unverified or left out.

Where this is, and how it relates to the 94158 briefing. This briefing covers the whole City and County of San Francisco, which is also the Climate Impact Lab impact region, so the mortality geography and the place match. A companion briefing covers ZIP 94158 (Mission Bay, Dogpatch, and east Potrero Hill) in more detail. Its actions and funding sit *inside* this one; do not add the two envelopes together.

What the forecast says. NOAA's Climate Prediction Center (8 October 2026) has an El Niño Advisory in force and calls a strong-to-very-strong El Niño likely through January–March 2027. It gives the California coast 50–60% odds of a wetter-than-normal winter. That is likely, not certain: the very strong 2015–16 El Niño left San Francisco at about 98% of normal rain, and the city's two wettest days came in a weak El Niño (November 1994) and a La Niña (31 December 2022). Prepare for atmospheric rivers in any winter; this one makes them more likely.

WHAT THIS CITY SHOULD DO

RANKED

Actions are ranked by expected lives saved per unit of effort before the peak. The backing lenses are shown on each action, and the RACI chart below assigns each one. Almost every action here changes a trigger or a list inside a program the city already runs. That is the finding: San

* Suggested formulation: check with a health worker before sharing.

Francisco has the programs, and it has the money ($832.7M for homelessness this year). What it lacks is the commitment to act before anyone has judged whether the warning is serious.

1. Commit in writing that an NWS Flood Watch or an AR3+ atmospheric-river forecast, on its own, opens the Emergency Operations Centre, deploys the flood barriers, and activates storm-night shelter

PARTIALLY CONFIRMED | EMERGENCY MGR · SKEPTIC · HISTORIAN · PUBLIC HEALTH | INVENTORY: NOT QUERYABLE BY CITY

OWNER	BY WHEN	COST	PREVENTS
Department of Emergency Management (DEM), SFPUC, Public Works, and HSH; the Board of Supervisors gets the commitment on the record at a November hearing	15 Nov	Staff time, plus overtime for false alarms (not quantified)	A repeat of 31 Dec 2022: barriers down and the EOC late while 5.5 inches fell

On 31 December 2022 the city expected about 0.75 inches. The NWS had forecast up to 3.4 inches over several days and issued a flood watch days ahead, and about 5.5 inches fell, the city's second-wettest day on record. The city's winter storm plan treated less than an inch as no threat, the Emergency Operations Centre opened late in the morning, and the deployable barriers at 17th and Folsom were not up; residents said “the city failed to show up,” although barriers had gone up before storms in earlier years. SFPUC told reporters its deployment follows the forecast. The barriers exist (a $165k pilot since 2017); only the trigger needs to change. Name the barrier sites in the commitment: 17th and Folsom, 15th Avenue and Wawona, and the Bayview and Islais Creek low points. The CW3E atmospheric-river scale (AR1–AR5, Ralph et al. 2019) is published for every forecast storm.

* Suggested formulation: check with a health worker before sharing.

Corrections: the late EOC opening and the “under one inch” threshold are from SF Public Press and were confirmed in the companion 94158 run; the flood watch's exact issue time was not confirmed. The Bayview red-tagged homes (Le Conte and Meade) and standing groundwater near Candlestick reported by one lens were not verified. No published after-action report and no written DEM or SFPUC trigger were found, so the city may already have changed its practice. Ask first. The SFPUC “forecasts decide” quote was seen only in a search summary.

2. Open storm-night shelter that people will actually use: published triggers, 24-hour intake that accepts pets, partners, and belongings, sites in Bayview and on the west side, a written pause on enforcement, and a safe place to park

PARTIALLY CONFIRMED | EMERGENCY MGR · SKEPTIC · PUBLIC HEALTH · PHILANTHROPIST · FRONTLINE | INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
HSH (accountable); Episcopal Community Services (ECS) runs Sanctuary and the Interfaith Winter Shelter; Healthy Streets Operations Center (HSOC) and SFMTA for the enforcement pause; outreach teams	**Triggers and sites by 15 Nov; outreach goes out before each storm's peak**	**Surge nights beyond the city's own at ~$126 per occupied bed-night; a new parking site is not costed**	**Cold, wet exposure among ~3,400 unsheltered people and those living in vehicles**

HSH's Cold & Wet Weather Protocol exists and ran about ten nights last winter at Next Door (1001 Polk) and Sanctuary (201 8th St). Its numeric triggers are unpublished, its sites are downtown, and nothing pauses enforcement while it is open. In the January 2023 storms the barriers to using shelter were documented: walk-up intake ran 4pm to 10pm on a first-come basis, and the city's street operation engaged 160 people over two weeks, of whom fewer than half accepted, with people citing no children allowed, chairs instead of beds, and fear of losing belongings. Since the oversized-vehicle ban took effect on 1 November 2025, more than 360 vehicles have been towed and about 1,100 citations issued (SF Standard, September 2026), and **no city safe-parking site is operating**: the Candlestick Vehicle Triage Center closed in March 2025 after costing more than $15M. Non-congregate shelter does change behaviour: in San Francisco's shelter-in-place hotels, emergency visits fell from 3.72 to 1.64 per person-year (Abbs et al., *J Urban Health* 2023; before-and-after, no control group).*

* Suggested formulation: check with a health worker before sharing.

Corrections: the “137 of 250 beds claimed” figure was not in its cited source and has been dropped. The 4–10pm intake is Mission Local's reporting, not SF Standard's. The “169 tows” figure was out of date. The Interfaith Winter Shelter's 2026–27 season had not been announced as of 9 October. No study shows that storm-night shelter reduces deaths; this rests on the documented gaps, not an effect size. A new parking lot before the peak is the least feasible part of this action.

3. Before December, list who would be stranded on an upper floor in an outage, and knock on those doors within hours of one

PARTIALLY CONFIRMED | FRONTLINE · EMERGENCY MGR · SKEPTIC · PUBLIC HEALTH | INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Department of Disability and Aging Services (DAS), with DEM; floor captains recruited through the SRO collaboratives (Chinatown CDC, Chinese Progressive Association, Mission Action, SOMCAN); SF Fire Department and NERT volunteers	**Lists by 1 Dec; checks within 6–12 hours of an outage**	**Staff and volunteer time; $100k–200k if funded through community groups (see funding)**	**Deaths among power-dependent and immobile residents in multi-day outages**

On 20 December 2025 a fire at PG&E's 8th and Mission substation cut power to about 130,000 customers. Most were restored within hours, but about 17,000 were still out at noon the next day, and KTVU reported elderly residents preparing for a fourth day without power, with wheelchair users unable to reach their floors. Project Open Hand served nearly double its usual clients, and DAS ran wellness checks and brought in Meals on Wheels. HHS emPOWER (queried live, 9 October 2026) counts **4,659 Medicare beneficiaries in San Francisco who use power-dependent equipment**, including 810 on dialysis and 735 on oxygen. Elevators are already failing: the Fire Department answered 1,764 elevator calls this year by late July against 1,277 in all of last year, concentrated in the Tenderloin, SoMa, and Chinatown, and one elevator at 845 California had been out 92 days. Use paper notices and a knock, because phone networks are unreliable when the power is out. The frontline lens put it plainly: in this city the storm usually doesn't kill you; the elevator does.*

Corrections: the “3–4 day” outage applied to a tail of customers, not the 130,000. No study measures how much wellness checks reduce outage deaths. The isolation risk factors one lens cited (living alone OR 2.3, not leaving home daily OR 6.7, top floor OR 4.7; Semenza et al., *NEJM* 1996) come from the 1995 Chicago *heat* wave, so they are an analogue only. The 2025 Iberian blackout raised deaths by about

* Suggested formulation: check with a health worker before sharing.

167 in Spain, mostly among women 85 and older, in the two days after power returned (Konstantinoudis & Riou, *Nat Commun* 2026). Medical Baseline gives notice, not priority restoration. Claims not reached: a ventilator patient's hospitalisation, PG&E's notification time to DEM, and an undelivered $10M SRO elevator pledge.

4. Put carbon monoxide and generator warnings, and a buddy phone tree for isolated seniors, into the languages San Francisco actually speaks

PARTIALLY CONFIRMED | PUBLIC HEALTH · FRONTLINE · SKEPTIC | INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
DEM and DPH for messages (Chinese, Spanish, Filipino, Vietnamese, Russian); Self-Help for the Elderly's centres for the Cantonese phone tree; SF Fire Department for alarms	Messages and phone trees before the first king-tide window, 23 Nov	Translation and printing; alarms for vehicle dwellers $5k–10k (assumed)	Carbon monoxide deaths from generators, grills, and heaters used indoors during outages

In New Mexico, CO detectors could have prevented about half of 136 unintentional, non-fire CO deaths (Yoon et al., *JAMA* 1998). In New York, a four-hour outage affecting at least 20% of a community raised the odds of a child's CO emergency visit by 150% or more that day (Northrop et al., *Pediatrics* 2025). Two men died at Sea Ranch in January 2023 with a generator running in a hallway. The callers in the phone tree check on isolated older residents in the Sunset and Richmond within 12 hours of a storm warning or an outage, and ask about the heater and the generator.*

* Suggested formulation: check with a health worker before sharing.

Corrections: Yoon's estimate is about half of *all* the deaths studied, not half of those asleep; it is a model, not a trial. The limited-English share and its language breakdown, the share of older adults living alone, and the claim that no Chinese-language NERT class exists could not be read from their sources and are not used. The five languages are the ones San Francisco's Language Access Ordinance names.

5. On storm and outage nights, carry naloxone and "never use alone" outreach into SROs and shelters, and watch the near-real-time overdose signal for a week after each storm

PARTIALLY CONFIRMED · ADD-ON | PUBLIC HEALTH | INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
DPH (DOPE Project, Naloxone Clearinghouse) and the Fire Department's EMS data	Built into the first activation	Within existing DPH supply	Overdose deaths, which dominate deaths among unhoused San Franciscans

Unhoused San Franciscans were 16 times as likely as housed residents to die suddenly (UCSF, *JAMA Internal Medicine* 2023, 868 autopsied sudden deaths over eight years), and overdose was the largest cause. Citywide overdose deaths were 810 in 2023, 633 in 2024, and 624 in 2025 (preliminary). Track EMS naloxone runs and emergency-department signals for days 0–7 after each storm, because Medical Examiner reports lag by one to three months and cannot act as a trigger.

* Suggested formulation: check with a health worker before sharing.

Corrections: **no study links rain or storms to overdose deaths**; the only weather link found is to sustained cold near 0°C, which San Francisco rarely reaches. This is a year-round program targeted at storm nights, not an El Niño measure in its own right. A daily excess-mortality feed from the Medical Examiner, proposed by one lens, is not feasible before the peak.

6. At South Ocean Beach, pre-arrange the emergency permit and a sand contract that fits the approved project, survey the bluff weekly, and close the beach and the Great Highway Extension on big swell at king tide

PARTIALLY CONFIRMED | HISTORIAN · SKEPTIC | INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
SFPUC (Ocean Beach Climate Change Adaptation Project); Public Works; Recreation and Park and SFMTA for closures; California Coastal Commission for the emergency permit	Permit and contract arranged by 1 Dec; first king-tide window 23–26 Nov	Not quantified	Loss of the bluff over the 14-ft Lake Merced Tunnel, and people swept off the beach

The bluff south of Sloat has failed in El Niño winters before. In January 2010 (a moderate El Niño), it had retreated more than 70 feet against its 2007 position, and waves came within about 20 feet of the Lake Merced sewage tunnel. Each emergency has been answered with rock. The long-term fix (permitted by the Coastal Commission on 14 November 2024) has slipped: SFPUC now expects construction to begin in late 2029 and take about four years. Sandbags went in along the southern frontage in spring 2025, and the permit requires annual monitoring, but no numeric trigger distance has been published. Add weekly surveys with a trigger distance, and arrange the sand and sandbag contract in advance so the emergency response fits the approved design rather than adding a fourth pile of rock. The NOAA king-tide windows are 23–26 November, 22–25 December, and 20–23 January.

* Suggested formulation: check with a health worker before sharing.

Corrections: the Great Highway Extension (Sloat to Skyline) is still open to traffic; Proposition K and Sunset Dunes covered the Upper Great Highway north of Sloat. The 2010 retreat was over 70 ft, not “up to 40 ft.” The 1997, 1998, and 2010 revetment sequence, and a nine-month, ~$5M lane closure, were not verified. No evidence was found that the emergency rock itself worsened erosion.

7. Use the money that moves before a flood: SFPUC mails its flood-risk parcels about the Floodwater Grant, landlords give the flood disclosure the law already requires, and households join PG&E's REACH queue now

PARTIALLY CONFIRMED | ECONOMIST (AS CORRECTED) | INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
SFPUC (grant); Rent Board and community groups (disclosure); PG&E (REACH, Match My Payment)	Mailers and REACH applications in October and November	Paid by SFPUC (up to $100k per property) and PG&E ($50M 2026 pledge)	Sewer-backup losses that nobody insures, and households entering storm season in arrears

Flood risk in San Francisco is held almost entirely by owners and renters. OpenFEMA shows **262 flood insurance policies in force citywide** on 1 January 2025, and 147 claims since 1978 totalling $2.66M. The 31 December 2022 storm produced 4 paid claims ($443,190). The SFPUC Floodwater Grant pays up to 100% of eligible costs, to $100,000, but only for properties with a prior flood or backflow and only if the agreement is signed before work starts, so it is a 2027 fix for most. California law (Gov. Code §8589.45) has required residential leases since July 2018 to disclose whether a property is in a flood hazard area. PG&E's REACH pays up to $800 with a disconnection notice, and Match My Payment up to $1,000 (to 400% of the poverty level).

* Suggested formulation: check with a health worker before sharing.

Corrections: matching 311 callers or NFIP claimants to addresses for outreach is not usable (Privacy Act and non-public 311 data), so SFPUC should mail its own 100-Year Storm map parcels. SFPUC already runs webinars and helps applicants in Chinese, Spanish, and Tagalog through 311. San Francisco's own 2019 flood-map disclosure ordinance exists, but its code citation could not be confirmed. The plumbing code's backwater-valve rule applies only where a fixture sits below the upstream manhole rim. The OpenFEMA v2 endpoints are deprecated on 15 October 2026.

8. Target the tree-hazard inspections that already happen at the blocks where people walk and sleep

PARTIALLY CONFIRMED | EMERGENCY MGR · PUBLIC HEALTH | INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Public Works Bureau of Urban Forestry; Recreation and Park for park trees	1 Dec	Existing crews	Deaths from falling trees in wind storms

On 21 March 2023 falling trees killed two people in San Francisco, on Post Street and at a bus stop on Lincoln Way, and about 700 trees or limbs came down (a record 662 calls to 311). Public Works already prunes on a cycle under StreetTreeSF; this action reorders that work toward mature trees on high-traffic sidewalks, near encampments, and in the Sunset grid.

No evidence was found that pre-storm pruning reduces deaths. A citywide sweep in two months is not feasible; a targeted one is.

9. On Treasure Island, keep a list of households with powered medical equipment or refrigerated medicine, and check the supply caches and shelter-in-place plan

PARTIALLY CONFIRMED | EMERGENCY MGR · FRONTLINE | INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Treasure Island Development Authority (TIDA) and DEM; SFPUC is the island's utility; a resident organiser keeps the list	1 Dec	Low	Medical emergencies when the island loses power and access

* Suggested formulation: check with a health worker before sharing.

About 2,800 people live on Treasure Island. Its one road link is a causeway to Yerba Buena Island and the Bay Bridge ramps, and SFPUC, not PG&E, supplies its power. The island has several outages a year.

Corrections: a ferry gives a second way off, though it may stop in storms, so “one causeway” overstates the isolation. No documented closure or isolation event, existing TIDA or DEM plan, or king-tide exposure record was found.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	DEM	HSH	ECS	SFPUC	Public Works	DPH	DAS	SF Fire / NERT	TIDA	PG&E	Community groups	Board of Supervisors
1. NWS watch triggers EOC, barriers, and shelter	A/R	R		R	R			C	I		I	C
2. Shelter people will use; enforcement pause; parking	C	A/R	R		C	C					R	C
3. Upper-floor outage list and door-knocks	C					C	A/R	R		I	R	
4. Multilingual CO warnings; senior phone tree	A/R					R	C	R			R	
5. Storm-night naloxone; overdose signal	I	C	R			A/R		R			R	
6. South Ocean Beach permit, surveys, closures	C			A/R	R			C				I
7. Grant mailers; lease disclosure; REACH				A/R			C			R	R	I
8. Targeted tree inspections	C	C			A/R							I
9. Treasure Island list and caches	R			C			C		A/R		R	

What the chart shows. **DEM is accountable for the first action but cannot deliver it alone**: the barriers belong to SFPUC and Public Works, and the shelter to HSH. A written commitment only binds if each of the three signs it. **The enforcement pause in action 2 has no accountable owner in this chart**, because HSOC and SFMTA, not HSH, run the clearings and tows; the Board of Supervisors is the lever.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by judgement, because no published study gives lives saved per dollar for cold or wet shelter in a US city. The envelope is bounded by what the city can absorb before the peak. San Francisco does not need donor money to build shelter: HSH's FY26–27 budget is $832.7M, with $377.3M for shelter and crisis over two years, but **no dedicated winter or inclement-weather line**. What a donor can buy is a lower trigger.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$253k	$605k	$2.58M	Fungibility: whether HSH would have paid for these nights anyway

Floor: extra activation nights 200 beds × 10 nights × $126.25 = $252,500. **Target:** 200 × 20 × $126.25 = $505,000 + wellness checks $100,000 = **$605,000**. **Ceiling:** 200 × 40 × $126.25 = $1,010,000 + Interfaith Winter Shelter backstop 80 beds × 126 nights = 10,080 bed-nights × $126.25 = $1,272,600 + wellness checks $200,000 + pooled fund $100,000 = **$2,582,600**. **The target fell 29% in verification**, from $855k: the $250k gift to the SF Foundation's pooled Rapid Response Fund was cut to a ceiling-only $100k because it cannot be restricted to

* Suggested formulation: check with a health worker before sharing.

storms, and the wellness-check floor was set to $0 because the work is uncosted and DAS may already fund it. The 200-beds-a-night capacity and the night counts are assumptions; $126.25 is the SF Controller's FY23 *average* cost per occupied bed-night, not the marginal cost of a surge night. The ZIP 94158 envelope ($191k / $489k / $1.26M) sits inside these figures.

RECIPIENT	USE	AMOUNT (TARGET)	MECHANISM	BASIS	VERDICT
Episcopal Community Services (2024 revenue $67.7M; runs Sanctuary under a $40.9M HSH contract)	Action 2: Cold & Wet nights beyond the city's own activations	$505,000	Restricted grant released per extra night HSH opens; conditional on HSH lowering its trigger	200 beds × 20 nights × $126.25. Last winter ran ~10 city-funded nights	PARTIALLY CONFIRMED
Self-Help for the Elderly (FY25 revenue $36.4M) and Chinatown CDC ($30.3M)	Actions 3 and 4: outage wellness checks and phone trees for isolated older residents	$100,000	Direct restricted grants	Uncosted judgement; check first whether DAS already pays for check-ins	PARTIALLY CONFIRMED
ECS, Interfaith Winter Shelter 2026–27	Action 2: a full season if the city does not fund it	$0 (ceiling $1.27M)	Contingent backstop	HSH has funded ECS for this every year; the season is unannounced. Congregation sites likely cost less than $126.25	CONTINGENT
SF Foundation Rapid Response Fund	General community response	$0 (ceiling $100k)	Unrestricted pooled gift	Cannot be restricted to storms	CEILING ONLY
Flood barriers and backwater valves	—	$0	Not funded	SFPUC already pays 100% of eligible costs (action 7)	ALREADY FUNDED

The funder's one thing: make the money conditional on HSH lowering its activation threshold, and pay per extra bed-night actually used. If the city would have opened those nights anyway, the gift only replaces city spending. Confirm with HSH that the extra nights and the Interfaith season are unfunded before writing a cheque.

* Suggested formulation: check with a health worker before sharing.

DO NOT DO THIS CONTRAINDICATED

Advice that circulates for this hazard and would cost lives, money, or trust here.

Bind a satellite-triggered municipal flood policy this winter

CONTRAINDICATED

One lens proposed copying Fremont's parametric flood policy, which pays $200k or more when satellites see at least 0.58 square miles of contiguous flooding (Floodbase; premium about 20% above Fremont's traditional policy; as of March 2025 it had never triggered). San Francisco's flooding is the wrong shape for that trigger. On 31 December 2022 it came as short, block-by-block sewer backups that drained within hours, and it probably would not have produced a contiguous area a satellite could see. A policy bought in December, with a strong El Niño already public, would also be priced for it. Money spent on a premium that does not pay is money not spent on action 2.

Instead: get a quote and a basis-risk analysis for 2027–28, against SFPUC's own flood records, and fund the before-the-flood measures in action 7 now.

Clear encampments or tow lived-in vehicles around a storm

DO NOT DO

KQED documented tent seizures before storms in November 2024, and advocates report clearings before the January 2023 rivers. A clearing on a storm day removes the only protection a person has, and scatters people away from the outreach workers who know where they are. With no safe-parking site open, a tow in a storm week puts a household on the street.

* Suggested formulation: check with a health worker before sharing.

Instead: a written enforcement pause whenever the Cold & Wet protocol is active (action 2).

Tell SRO and high-rise residents to “shelter in place” without knowing who cannot use the stairs

DO NOT DO

“Shelter in place” assumes a working elevator, a charged phone, and someone to bring food. In December 2025 wheelchair users could not reach their floors, and elevator calls this year are running at about double last year's. Advice that assumes those things leaves the most immobile residents alone on the fourth day.*

Instead: the upper-floor list and door-knocks in action 3. Also do not tell people that Medical Baseline gets their power back first, or that a winter shutoff moratorium protects them; the moratorium applies only below 32°F.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Jan 1982	Bay Area storm; ENSO-neutral (ONI −0.1)	33 Bay Area deaths, 25 from landslides (mostly outside SF)	USGS landslide-warning research
1982–83	Very strong El Niño (ONI 2.14); SF season 38.17 in	High water and surge in the Bay; SF toll not found	—
1994	Weak El Niño; SF's wettest day, 5.54 in (5 Nov)	Ocean Beach bluffs lost 30–40 ft in the 1994 storms	Rock revetments at Sloat
1997–98	Very strong El Niño; SF season 47.19 in; wettest February	Landslide damage $4.1M in SF, $55M in San Mateo County, $158M across the region (USGS)	—
2009–10	Moderate El Niño	South Ocean Beach bluff retreated 70+ ft since 2007; waves within ~20 ft of the Lake Merced Tunnel	More rock; SPUR Ocean Beach Master Plan (2012) proposes managed retreat

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2015–16	Very strong El Niño (ONI 2.59)	SF about 98% of normal rain; sandbag pop-ups; expanded shelter	—
31 Dec 2022	SF's second-wettest day, 5.46 in; La Niña	EOC late; 17th & Folsom barriers not up; 4 paid NFIP claims ($443,190)	17th & Folsom stormwater project finishing late 2027
Jan 2023	Atmospheric-river series	Fewer than half of 160 people engaged accepted shelter	No published after-action report found
21 Mar 2023	Wind storm	2 killed by falling trees; ~700 trees or limbs down	—
2024	Coastal permit for South Ocean Beach (14 Nov)	—	Construction now expected from late 2029
20 Dec 2025	8th & Mission substation fire	~130,000 customers out; some older residents into a fourth day	Supervisors' hearing on PG&E

The recurrence nobody has noticed: at Sloat, the emergency response is the only part of the plan the city has ever built. Rock went in after the 1990s storms and again after 2010, while the permanent fix has been a few years away since 2012. If the bluff fails this winter, the emergency response will again become the plan, unless it is designed now to fit the 2029 project.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Buy a municipal parametric flood policy** (the Economist) versus **“the trigger will not fire for San Francisco's kind of flooding”** (verification). Resolved for this winter as contraindicated. What would settle it: run Fremont's trigger against SFPUC's record of the 31 December 2022 flood.
- **“The storm kills through the elevator, not the water”** (Frontline) versus **“the storm kills through overdose, indoors and alone”** (Public health). Both are plausible, neither is measured, and the actions (3 and 5) do not compete. What would settle it: linking Medical Examiner deaths to outage and storm dates.

* Suggested formulation: check with a health worker before sharing.

- **A safe-parking site for vehicle dwellers** (Emergency mgr, Frontline) versus **the Candlestick site's cost of more than $15M, about $140k per space** (verification). Unresolved; a cheaper short-term lot was not found.
- **Hypothermia deaths in San Francisco are near zero** (one undated count) versus **San Francisco is among the highest-rate large California counties** (KFF Health News). Unresolved; recent Medical Examiner exposure-death counts were not retrieved.

WHAT NOBODY COVERED — BLIND SPOT

Hillsides and the waterfront at king tide. No lens found a sourced landslide record for San Francisco's own steep neighbourhoods (Telegraph Hill, Twin Peaks, Diamond Heights, Bernal Heights), although one cited a 2025 Telegraph Hill rockfall that could not be confirmed. And no lens examined the Embarcadero and the seawall when a storm falls on a king tide, which is where the Port's $425M seawall bond is being spent. Both deserve a check with Public Works' geotechnical staff and the Port's Waterfront Resilience Program before December. Separately, one lens reported stalled driverless cars blocking responders in the December 2025 outage; that was not verified.

HOW TO READ THIS — METHOD

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale. Here that region is the whole city. The El Niño projection covers *heat only*; flood, cold, wind, outages, carbon monoxide, erosion, and overdose are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** The request named “the city of San Francisco”. Climate Impact Lab impact region, as the report records it: San Francisco County, CA, *inferred* rather than resolved on the Impact Map; the city and county are coterminous. Actions are citywide and therefore less specific than a neighbourhood briefing; the companion ZIP 94158 briefing shows what one neighbourhood looks like in detail.

* Suggested formulation: check with a health worker before sharing.

- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) researched the city independently and in parallel, from web sources, to a fixed brief. Each was given the facts already verified in the 94158 run, to avoid re-checking them, and pointed at the parts of the city outside Mission Bay.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias: four lenses converged on pre-set triggers, and their evidence is San Francisco's own 2022 record, so it stands. Contradictions were kept. Actions were ranked by expected lives saved per unit of effort before the peak, given payers or marked as within existing budgets, and reconciled against the funding placements.
- **Verification.** Seven verifier agents checked 12 recommendations, the historical record, the financial instruments, and the funding placements on evidence, transferability, and feasibility. Verifiers queried the HHS emPOWER county layer, OpenFEMA NFIP policies and claims, and IRS filings on ProPublica directly. The envelope was re-summed. Facts verified in the 94158 run (NOAA outlook, ONI values, HSH protocol, SFPUC grant terms, PG&E programs, overdose figures) were reused, not re-checked.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not queried in this run. One verifier found it can be filtered by country, sector, and hazard but not by city, and it rendered 0 entries when fetched. A 94158 verifier who did query it found only a Canadian cold-weather alert program and US heat advisories as the nearest entries. Every action is tagged accordingly.
- **Health wording.** Passages where a wording error could hurt someone (cold-weather shelter, medical equipment in an outage, CO alarms, and sheltering in place) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so. Assumed capacities and night counts are labelled.
- **Limits.** Several verifiers exhausted their web-search budget and checked by fetching primary pages; American Legal, leginfo, and Justia returned 403, and the Controller's report index returned 500. As a result: the city reserve balances and their rules are unverified and the

* Suggested formulation: check with a health worker before sharing.

reserve action is not used; the SF flood-disclosure ordinance citation is unconfirmed; the limited-English language shares are not used; the Telegraph Hill rockfall, Bayview red-tags, a 2004 flood settlement, and a nine-month Great Highway lane closure were not verified and are not used. The panel and verifiers are AI research agents, and no one in San Francisco was interviewed.

References · verified by Phase 5 unless marked. NOAA CPC ENSO discussion, 8 Oct 2026 (verified) · NOAA CPC long-lead outlook (verified) · NOAA CPC ONI table (verified) · SF seasonal rainfall (verified) · Mission Local, 17th & Folsom forecast and barriers (verified) · Mission Local, "the city failed to show up" (verified) · SF Public Press, 2022–23 emergency response (verified in 94158 run) · CW3E atmospheric-river scale (verified) · HSH inclement weather protocol (verified) · Interfaith Winter Shelter (verified) · Mayor's release, 2025–26 winter shelter (verified) · Mission Local, January 2023 storm shelter (corrected) · SF Public Press, shelter hurdles (verified) · SF Standard, RV enforcement, Sep 2026 (verified) · KALW, Vehicle Triage Center closure (verified) · Abbs et al. 2023, SF shelter-in-place hotels (verified) · KQED, tent seizures before storms (verified) · 2026 Point-in-Time Count, preliminary (verified) · T&D World, December 2025 substation fire (verified) · CBS, December 2025 outage (verified) · KTVU, outage and older residents, 22 Dec 2025 (verified) · DAS Commission minutes, Jan 2026 (verified in 94158 run) · SF Standard, elevator calls, Jul 2026 (verified) · HHS emPOWER county layer (queried 9 Oct 2026) · Semenza et al., *NEJM* 1996 (verified; heat analogue) · KQED, supervisors probe PG&E (lens) · Yoon et al., *JAMA* 1998 (corrected) · Northrop et al., *Pediatrics* 2025 (verified in 94158 run) · KTVU, Sea Ranch generator deaths (verified) · Self-Help for the Elderly, IRS filings (corrected) · Chinatown CDC, IRS filings (verified) · UCSF, sudden death among unhoused people (verified) · OCME overdose reports (verified) · Cold and overdose, *Epidemiology* 2019 (verified) · SFPUC Ocean Beach Climate Change Adaptation Project (verified) · SFPUC Ocean Beach update, July 2025 (verified) · SPUR, El Niño and Ocean Beach (lens) · SPUR Ocean Beach Master Plan (verified) · USGS MF-2325, 1997–98 landslide damage (verified) · USGS PP 1434, January 1982 storm (verified) · OpenFEMA NFIP policies (queried) · OpenFEMA NFIP claims (queried) · SFPUC Floodwater Grant program (verified) · SFPUC Floodwater Grant terms, Nov 2025 (verified) · CA Gov. Code §8589.45 (from recall; page returned 403) · PG&E REACH (verified) · PG&E Match My Payment (verified) · Floodbase, Fremont municipal policy (verified) · SF Office of Risk Management (verified) · SF Standard, March 2023 trees (verified) · Mission Local, record 311 calls (verified) · NBC Bay Area, March 2023 deaths (verified) · SF Public Works, StreetTreeSF (verified) · Treasure Island overview (secondary) · HSH budget FY2026–28 (verified) · SF Controller shelter assessment, Dec 2024 (verified) · Episcopal Community Services, IRS filings (verified) · SF Foundation Rapid Response Fund (verified) · Port of SF Waterfront Resilience Program (lens)
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory
Generated 9 October 2026 for San Francisco, California. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

* Suggested formulation: check with a health worker before sharing.